

Review article

Systematic review of data-centric approaches in artificial intelligence and machine learning

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ABSTRACT

Artificial intelligence (AI) relies on data and algorithms. State-of-the-art (SOTA) AI smart algorithms have been developed to improve the performance of AI-oriented structures. However, model-centric approaches are limited by the absence of high-quality data. Data-centric AI is an emerging approach for solving machine learning (ML) problems. It is a collection of various data manipulation techniques that allow ML practitioners to systematically improve the quality of the data used in an ML pipeline. However, data-centric AI approaches are not well documented. Researchers have conducted various experiments without a clear set of guidelines. This survey highlights six major data-centric AI aspects that researchers are already using to intentionally or unintentionally improve the quality of AI systems. These include big data quality assessment, data preprocessing, transfer learning, semi-supervised learning, machine learning operations (MLOps), and the effect of adding more data. In addition, it highlights recent data-centric techniques adopted by ML practitioners. We addressed how adding data might harm datasets and how HoloClean can be used to restore and clean them. Finally, we discuss the causes of technical debt in AI. Technical debt builds up when software design and implementation decisions run into “or outright collide with” business goals and timelines. This survey lays the groundwork for future data-centric AI discussions by summarizing various data-centric approaches.

1. Introduction

Machine learning (ML) is, and has always been, about data. However, data science teams have only recently moved their focus to data owing to the introduction of strong push-button models. This method, known as data-centric artificial intelligence (AI), entails constructing intelligent systems using high-quality data with an emphasis on ensuring that the information clearly communicates what the AI needs to learn. Furthermore, it entails the iteration and collaboration on the data needed to develop AI systems programmatically. Why is the industry placing greater emphasis on data centrality? A reasonable starting point is to contrast data-centric AI with model-centric AI, which has been the focus of ML research for many years. Model creation has traditionally emphasized data science and ML teams with iterations of feature engineering, algorithm design, and bespoke model architecture. They considered the data as static items, and most teams concentrated on the model (Chakraborty and Krishna, 2014).

However, as models have become more sophisticated and push-button, AI teams have realized that focusing on data iteration is just as important, if not more so, for developing and deploying high-accuracy models successfully and efficiently. ML models have become increasingly complicated and opaque in recent years, necessitating significantly larger amounts of training data. In addition, data have evolved into a useful interface for working with subject matter experts and transforming their expertise into software. Finally, data-centric AI enables a higher level of model accuracy than was previously feasible using only model-centric techniques.

Structured, semi-structured, and unstructured data are the three categories of data sources that can be classified (Eberendu, 2016). Unstructured data account for approximately 80% of all global data, whereas structured data account for only 20% (Kumar et al., 2020).

Large volumes of data can result in complexities. Consequently, the use of simple data analysis methods has dwindled. There are simply too many semi-structured and unstructured data for processing. Researchers

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have coined the term big data analytics (BDA) to describe large and complicated datasets. “Big data” refer to the ever-increasing amount of structured and unstructured data. Relational databases cannot be managed by simple data management tools. Therefore, BDA assembles various tools and methodologies to handle large and complex datasets.

Researchers have used several characteristics to develop a broad definition of “big data”. First, researchers established three characteristics of big data: quantity, speed, and variety. As big data grew in complexity, researchers began to incorporate more data characteristics to accurately characterize high-quality data. The V of big data refers to these features (Tabesh et al., 2019). Finally, some academics have specified up to 17 characteristics of big data based on the difficulties encountered when dealing with large amounts of data. We covered the ten V’s big data in the sections that follow (Gandomi and Haider, 2015; Oussous et al., 2017; Panimalar et al., 2017; Saggi and Jain, 2018).

The goal of data-centric AI is to continuously improve data quality. It undergoes several steps in the ML lifecycle after being generated and collected. An ML lifecycle using a data-centric approach is illustrated in Fig. 1. Once the data have been cleaned up, they are augmented and labeled before they are finally analyzed. Preprocessing is an important aspect of data-centric AI. To increase the accuracy, many trials were conducted using the data processed in the model training phase.

The major contribution of our proposed work is as follows:

- Summarizing the basic statistics on curating big and high-quality data to build a foundation for discoursing data-centric AI methods.
- Exploring all major characteristics of the data-centric AI approach and framework.
- Highlighting the most recent improvements in machine and deep learning based on data-centric approaches.
- Summarizing the state of data-centric AI as a guideline for future research.

The remainder of this paper is organized as follows: Section 2 highlights the previous studies that summarize data-centric approaches. Section 3 covers the research methodology and describes the exact process followed to collect relevant research papers and materials for the survey. Section 4 summarizes the major aspects of the data-centric approaches in ML. It covers big-data quality assessment, various preprocessing techniques, transfer learning, semi-supervised learning, and

machine learning operations (MLOps). This section reviews the recent data-centric techniques proposed by ML practitioners. Section 5 concludes the survey by summarizing all the data-centric aspects and outlining future research directions.

2. Related work

Data-centric AI has not been widely reviewed. Few studies have examined individual techniques related to data-centric AI. A comprehensive review of this budding domain is lacking in academic literature. In this section, we highlight the efforts of AI researchers in reviewing this field. The work from Miranda (2021) presented a short review of data-centric ML closely related to our survey but on a smaller scale. The author listed the reasons for the shift in the AI approach from model centric to data centric, including the democratization of deep learning (DL) and the introduction of software engineering practices in ML. This study identified seven data-centric AI techniques: transfer learning, multi-task learning, crowdsourced labeling, semi-supervised learning, weak supervision, active learning, and data assimilation. The first four techniques do not require high-domain expertise. The last three require high-domain expertise to improve the efficiency of the model by preparing high-quality data. The research concludes by mentioning data-centric AI as a promising field that has been around for the last three decades, but has been formally defined only recently.

The recent work from Ramponi and Plank (2020) highlighted the effectiveness of domain adaptation (DA) in natural language processing (NLP) and explored various model-centric, data-centric, and hybrid approaches in this field. The authors reviewed pseudo-labeling, data selection, and pre-training techniques for data-centric NLP. They highlighted future research directions in DA, such as creating comprehensive benchmark datasets for unlabeled domain adaptation.

Polyzotis et al. (2018) surveyed different data management tools and literature. This study focuses on three data management components: data understanding, data validation and cleaning, and data preparation. The authors focused on the challenges faced during each phase of the data management lifecycle and highlighted the methods for tackling them. For example, in understanding data, it is important to implement sanity checks to examine whether they are useful in the current pipeline. It is also important to analyze the data features for better understanding. For data validation and cleaning, this study reviewed methods that can

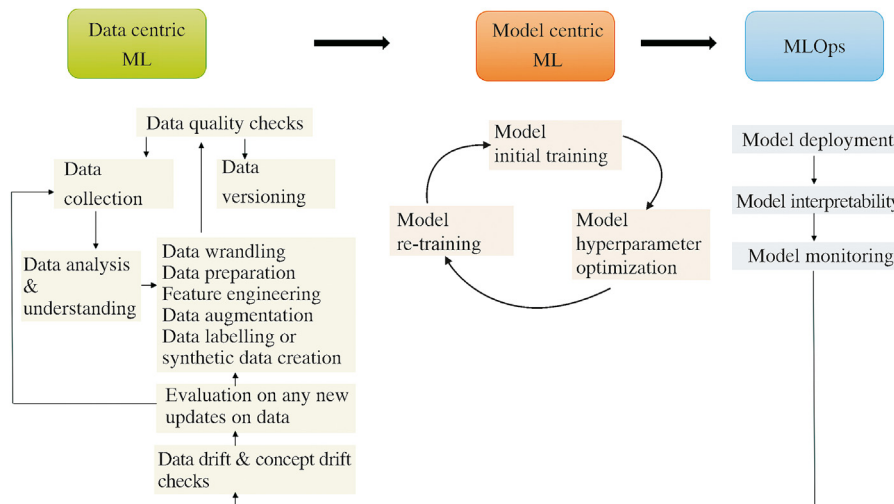


Fig. 1. Data-centric design lifecycle.

issue alerts to humans if there is a lapse in data validation. Finally, the authors conducted numerous feature engineering and data enrichment studies on data preparation. Bifulco et al. (2021) introduced CAIMANS, an intelligent system designed to assist enterprises in selecting artifacts of interest from the Web by checking their compatibility with a company's recorded expertise and background information. Consequently, CAIMANS uses a sophisticated crawler to scan the Web for relevant artifacts. A thorough validation of e-procurement use cases with real-world dimensions was conducted in collaboration with industry stakeholders. In the context of AI in human-centric applications, such as security, robustness, interpretability, and ethical (data and algorithmic) considerations (Ahmad et al., 2022). This study provided a detailed analysis of the existing research on the fundamental challenges of efficient implementation. In addition, this study investigated how these problems overlap and may lead to, or contribute to, the resolution of further issues. Researchers have explored the current limitations, issues, and future directions of research in various fields, as well as how they might fill the current gaps and develop more effective solutions. The extensive analysis presented in this paper provides a basis for future research on this topic.

Yang et al. (2022) presented a multi-factor attention strategy for balancing the impact of indiscriminate features on different prediction targets. First, the suggested parking availability prediction (PAP) system adopts an attention mechanism to improve the performance of parking availability prediction. Subsequently, the authors assessed the proposed technique using two large-scale primary datasets obtained from China and the U.S. Finally, the results demonstrated that PAP obtained superior prediction results for urban parking and truck parking prediction for five prediction timeslots. Even with insufficient training data, the proposed approach displayed superior generalizability and flexibility for different lot types. In addition, the Washington State Department of Transportation (WSDOT) selected and utilized the projected Smart Parking Data Management and Prediction (SPDMP) as a pilot testbed for truck parking information dissemination.

The studies reported by Crawshaw (2020) and Ruder (2017) present a detailed view of multi-task learning (MTL) techniques, which are classified into two groups: hard- and soft-parameter sharing. The authors reviewed the techniques from both groups and highlighted various architectures built on multitasking learning principles. MTL implicitly presents a data-centric approach by sharing parameters among various ML tasks to achieve better generalization. Parameter sharing allows data augmentation because one task can use data from another to improve the performance. The author observed that the MTL lacks theoretical literature to better understand MTL-based deep neural networks (Crawshaw, 2020).

Roh et al. (2019) surveyed a comprehensive list of data collection techniques, primarily data acquisition, data labeling, and improvement of existing data. For data acquisition, they reviewed data discovery, sharing, searching, augmentation, integration, generation, and crowdsourcing. For data labeling, the study extensively reviewed the literature on techniques such as self-labeling, graph-based label propagation, active learning, and weak supervision. Various data cleaning, re-labeling, and transfer learning techniques focused on improving data quality, and models from previous studies were also discussed. The authors also presented future research directions. Their observations indicated that data evaluation, data integration, and crowdsourcing techniques could be improved. Several intelligent algorithms for focused crawling of online artifacts that semantically match them against internal big data and knowledge sources have been proposed in the literature (Caruccio et al., 2021). The components of the crawlers were evaluated by integrating them into a system. The proposed method has been deployed and thoroughly evaluated in the e-procurement domain, and has gradually been extended to other areas, including robot programming and cloud service providers. Using healthcare data, Gordon et al. (2022) objectively evaluated freely available data profiling software tools. Initially, data profiling methods were examined using publicly available information

and datasheets to assess their capabilities. Using a synthetic dataset of one thousand patients and the associated data derived from a standard health data model, several healthcare data analysis tools were selected for further examination. The tools were evaluated based on their compatibility with the dataset.

Sarker et al. (2020) presented a framework based on data-centric NLP and ML to monitor the abuse of social-media-based prescription medicine (PM). The authors first reviewed 39 studies targeting social media-based PM abuse or misuse to analyze trends. The study found that previous techniques fell under three categories: manual, unsupervised, and supervised. Based on their findings, the researchers proposed a data-centric NLP framework to mine social media for PM misuse. The framework's data processing pipeline collects data from social media sites such as Twitter and filters them to remove irrelevant posts and noise. Once cleaned, the data are passed through NLP and ML systems for further processing. Additionally, the proposed framework can employ techniques such as chatter analysis and geotagged social media data to identify topics or content related to PM abuse.

Lwakatare et al. (2020) presented a systematic review of 72 research papers on large-scale ML systems. This study highlights many challenges and solutions related to ML workflow components, such as data acquisition, training, evaluation, and deployment. In addition, this study reviews the data quality, data dependency, sparse data, limited data access, difficult data exploration, data labeling, and feature-engineering problems in numerous large-scale ML systems related to the automotive, aviation, online services, telecommunications, medicine, and industrial automation industries.

Our study presents an end-to-end view of a data-centric approach to AI and ML. This study differs from the previous literature because it outlines all the major aspects of data-centric AI. The novelty of our work lies in the comprehensive approach adopted to review data-centric techniques. It includes summaries of state-of-the-art techniques related to data-centric AI, which presents a complete view of this developing domain.

3. Research methodology

This section describes the methods used to conduct this research review. 28 papers were selected from notable publishers such as IEEE Xplore, ACM, and Elsevier. Open-access research papers are also included in arXiv. Various search queries were used to filter the research papers related to data-centric AI approaches. Most research papers were collected from 2019 to 2022. However, if any notable related research was found, it was also added to this survey. Because data-centric AI is a relatively new field, academic research that specifically mentions data-centric AI is limited. Therefore, research material from high-quality websites was included in this survey. Table 1 presents a breakdown of the research papers collected from various databases and categorized according to data-centric AI aspects.

4. Data-centric literature survey

This section presents a detailed review of the recent advances in data-centric paradigms. Researchers have presented various techniques and frameworks employing data-centric strategies to improve the performance of ML processes. This survey explored all major data-centric modeling aspects to present a comprehensive view of the current state of this field. Table 1 summarizes the research papers discussed in this study. If researchers are working in a specific area of the data-centric approach, this table will provide quick insights. What work has already been done in these areas and what methods/techniques have been adopted or implemented? This table effectively categorizes research on various aspects of improving ML performance through data quality improvement.

Table 1
Summary of data-centric artificial intelligence (AI) approaches in machine learning (ML).

Paper reference	Data-centric AI aspect	Techniques proposed	Results/future directions/limitations
Murphy (2019)	Big data to good data: good data vs. bad data assessment	A system that differentiates between various kinds of data.	It helps to identify datasets liable to cause damage and unbearable costs.
Tian et al. (2018)	Big data to good data: data quality assessment	Data quality monitoring system that develops data checkout rules on theoretical power data.	Data checkout rules improve the quality of the power supply service by refining the power data. The proposed system needs to be validated for real-time power data.
Juneja and Das (2019)	Big data to good data: data quality assessment	Pre-processing framework for monitoring data quality for a conceptualized weather forecasting system.	The proposed pre-processing techniques can theoretically improve the forecast accuracy of weather data. The framework needs to be validated for real-world data.
Taleb et al. (2018)	Big data to good data: data quality assessment	Quality evaluation model for UBD.	The proposed model presents a data quality report for UBD on which an initial UBD schema can be built. More experiments are needed to validate this model.
Taleb and Serhani (2017)	Big data to good data: data quality assessment	DQR framework to accurately identify the pre-processing activities for each data feature of a big data set.	After applying DQR, the quality of the data features is improved. Further experiments with more big datasets are needed to validate the framework.
Anik and Bunt (2021)	Big data to good data: data quality assessment and explanation using explainable AI	Data-centric explanations for training datasets.	Improved understanding and trust of the end-user on the ML systems. Real-world scenarios and end-user diversity are needed for comprehensive results.
Meng et al. (2020a)	Big data to good data: ML-based data fusion techniques	A review of various data fusion techniques that belong to three data fusion types: signal level, feature level, and decision level	Data fusion improves the quality of the datasets. More robust data fusion models must be built on large-scale systems.
Quan et al. (2020)	Data collection and labeling	Data-centric method to generate CG images to improve the generalization ability of CNN-based detectors.	CG images generated using a data-centric technique along with a novel CNN detector improved the results of the blind detection problem. Results could be further improved by improving the image generation formulation.
Wadekar et al. (2021)	Data collection and labeling	Novel data collection strategy for autonomous racing and a novel model for steering and throttle prediction.	The proposed data collection technique shows promising results in steering and throttle prediction. More work is required for better validation. The study does not present a detailed literature review to understand the problem or its limitations in detail.
Sidiropoulos et al. (2020)	Data collection and labeling	RNN-based synthetic data generation for the KGSQA system.	The question-answer pairs generated using the proposed techniques perform better than SOTA methods.
Kim et al. (2020)	Data collection and labeling	Simulating LSBN for generating large-scale LSBN data.	Large datasets are generated in the simulated LSBN domain. Further experimentation on these datasets is required to consolidate results.
Bérard et al. (2020)	Data collection and labeling	Multi-domain and multilingual language translation model specifically for COVID-19 biomedical data.	The model presents comparable results with the previous SOTA.
Yoon et al. (2022)	Data collection and labeling	Consistent data labeling and removing noisy data from a biomedical dataset to improve the performance of the pre-trained BioBERT model.	BioBERT model showed improved results for the question-answering task. Further work can be done to validate the results.
Lee et al. (2021)	Data collection and labeling	Faster R-CNN model with RPN to improve the accuracy of detecting fire and smoke based on a novel data labeling approach.	Better results are achieved using a balanced labeling approach and further experimentation is needed to improve results.
Wang et al. (2020)	Data collection and labeling	LILA framework for synthetic label generation on incomplete labeled data.	The proposed framework generates favorable results. The work can be further extended to solve the class multi-label imbalance problems.
Zhang et al. (2020)	Data augmentation	GAN-based DADA technique to generate high-quality images for small labeled datasets.	DADA improves the performance of small labeled datasets for classification tasks. It can be extended to solve object detection tasks across various domains.
Mansourifar et al. (2019)	Data augmentation: oversampling technique	V-GAN model is proposed that can increase the dimensions of original training data to produce virtual big data.	V-GAN outperforms traditional GAN and SMOTE oversampling techniques to improve prediction results.
Zhang et al. (2022b)	Data augmentation	Horizontal and vertical circular shift data augmentation technique works well for preventing data loss in small images during the augmentation process.	The circular shift operation shows consistent improvement for CNN-based models for classification tasks. Further improvements can be done.
Cooney et al. (2021)	Feature engineering	Conceptual ML framework that uses economic, nutritional, and environmental datasets to optimize aquatic feed.	This framework can potentially result in sustainable and optimized aquatic feed compositions but needs to be validated practically.
Sharma and Liu (2020)	Feature engineering	Data-centric ML model for misbehavior detection in IoV.	The proposed framework gives favorable results. It must be tested with real-world datasets for further validation.
Punmiya and Choe (2019)	Feature engineering	Proposed GBTD for energy theft detection based on feature engineering.	The proposed model improves the performance and time complexity of theft detection. This work should be further tested with other datasets and models.
Huang et al. (2019)	Feature engineering	Proposed features engineering framework for earthquake prediction.	5 features extraction techniques are used. The resulting features will be used in the future to develop an ML earthquake prediction framework.
Alzahrani et al. (2019)	Feature engineering	Inception V3 CNN model with HOG and facial landmarks-based feature extraction technique to classify face shapes.	The proposed technique performs better than the previous SOTA. The work can be improved with better datasets and better feature extraction.
Sanjeeva (2018)	Data management	Proposed context of data storage and maintenance with challenges related to it.	The presented techniques guide the challenges and solutions in big data management and also a point of reference for future work

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Table 1 (continued)

Paper reference	Data-centric AI aspect	Techniques proposed	Results/future directions/limitations
Schiermeier et al. (2018)	Data management	Sponsorship and implementation of “open science” the push to make scientific research and data freely accessible.	The research can result in universal practice of data management by default with data plans
Sundarraj (2022)	Data management	Effective data management in businesses.	This research proposed the relationship between data management and customers' satisfaction for a successful business model.
Bossér et al. (2020)	Semi-supervised learning	Labeling techniques based on novel GPLA for unlabeled data.	The proposed technique improved the performance of ANN.
Chao et al. (2021)	Semi-supervised learning	Label large amounts of unlabeled SCADA data to test the performance of DNN.	DNN shows better results with more labeled data.
Shao et al. (2019)	Semi-supervised learning	DNN-based semi-supervised learning technique for sample labeling malicious communication.	The proposed technique has better labeling accuracy than K-NN and LPA.
Noorbehbahani and Saberi (2020)	Semi-supervised learning	A random forest-based novel technique for ransomware detection based on chi-squared or OneR feature selection.	The model outperforms supervised learners in this domain. More work can be done in terms of developing a semi-supervised feature selection technique for this problem
Zhang et al. (2020a)	Semi-supervised learning	Tri-CEAL to improve labeling for deep image classification tasks.	Improved labeling accuracy with Tri-CEAL. In the future, labeling accuracy can be improved.
Gururangan et al. (2020)	Domain adaptation & transfer learning	Improve RoBERTa with pre-training using four different domains.	RoBERTa's performance improved with pre-training. Data selection and domain adaptation strategies can be improved in the future.
Han and Eisenstein (2019)	Domain adaptation & transfer learning	AdaptaBERT for unsupervised language modeling.	AdaptaBERT performs better than other BERT variants in some cases. In the future, a single unsupervised language model can be developed for various domains.
Ben-David et al. (2020)	Domain adaptation & transfer learning	PERL model for unsupervised DA with MLM objective fine-tuning.	The proposed model outperforms on 22 binary sentiment classification tasks. This technique can be extended to other NLP tasks.
Nguyen and Chang (2021)	Domain adaptation & transfer learning	Predicting COVID-19 severity using data-centric pre-training and in the proposed hybrid convolution-attention neural architecture.	The proposed model performs better than conventional approaches. This work can be improved by experimenting with different self-attention modules.
Dou et al. (2019)	Domain adaptation & transfer learning	DAFE model with back translation for unsupervised DA of the neural machine translation task.	Improved results for neural machine translation. Different language datasets can be tested further to validate results.
Mäkinen et al. (2021)	MLOps	Survey of 331 data science experts to analyze the state of MLOps.	Different companies are using data and ML differently. Some are only using data, others are building ML models with that data. Only a handful have automated ML pipelines.
Zhou et al. (2020)	MLOps	Building an MLOps platform with existing ML tools to check its feasibility.	The developed MLOps platform is able to execute various ML models effectively.
Fursin et al. (2020)	MLOps	CodeReef open-source platform for ML SysOps.	This is a cross-platform tool that can automate ML workflows. It can be expanded by the open-source community.
Renggli et al. (2021)	MLOps	Discussed four data quality dimensions in MLOps workflows.	Provided tools to cater to each aspect of data quality along with future research direction in terms of ML-aware data quality.
Granlund et al. (2021)	MLOps	Integrating and scaling MLOps workflows across various organizations.	The solution is limited to the analyzed case studies. Broader experimentation is needed across organizations to identify similar workflow patterns.
Chen et al. (2022)	MLOps	Submarine framework to address various ML challenges.	The platform is lightweight and portable. It is easy to use for administrators and data scientists and provide efficient ML workflows.
Dilmegani (2021)	MLOps	Various tools and platforms for MLOps.	MLOps tools fall under three categories: data management, modeling, and operationalization.
Miranda (2021)	MLOps	Discussed MLOps landscape.	Various cloud-based and standalone specialized MLOps tools are listed.
Czakov (2020)	Experiment tracking	Discussed experiment tracking using Neptune platform.	Various aspects of experiment tracking are discussed. Experiment tracking tools like Neptune can save ML metadata effectively.
Schelter et al. (2017)	Experiment tracking	Lightweight ML experiment tracking platform.	The proposed platform can extract and store ML experiment data. The platform is limited to capturing physical data operations only. For logical data operations, the platform needs to be extended.
Chen et al. (2020a)	Experiment tracking	Discussion on MLflow tool and its features.	MLflow offers experiment tracking and many other features. The tool can be improved by incorporating solutions for various data challenges.
Nakkiran et al. (2021)	Can adding data hurt?	Deep double descent: where bigger models and more data hurt.	This framework helps identify certain regime where increasing train sample number can hurt data test performance.
Min et al. (2021)	Can adding data hurt?	Production of models that are robust to perturbations on the input data.	This research help mitigate the decrease in data standard accuracy. More experiments are need on this research for proof declaration.
Li et al. (2020)	Can adding data hurt?	Provable more data hurt in high dimensional least squares estimator.	This paper investigates the finite-sample prediction risk of the high-dimensional least squares estimator.
Rekatsinas et al. (2017)	Repairing data	Holistic data repairs with probabilistic inference.	The proposed method explain how data can be repaired with HoloClean driven by probabilistic inference.
Lee et al. (2021)	Repairing data	Explanation on data cleaning.	The section of this paper explained data cleaning as the initial stage of any ML project and its one of the critical process in data analysis. Some few procedures are required.
Bogner et al. (2021)	Data in deployment	A data-centric view of technical debt in AI.	The proposed technique explains that pushing to develop software very quickly can create long term maintenance costs and can sometimes be unbearable.

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Table 1 (continued)

Paper reference	Data-centric AI aspect	Techniques proposed	Results/future directions/limitations
Sculley (2022)	Data in deployment	ML and AI special ability to create large amount of technical debt.	The framework exposed how ML and AI can bring about huge technical debt.

Note: unstructured big data (UBD); data quality review (DQR); computer generated (CG); convolutional neural network (CNN); recurrent neural network (RNN); knowledge graph simple question answering (KGSQA); location simulating based social networks (LSBN); risk priority number (RPN); language-informed latent actions (LILA); generative adversarial network (GAN); deep adversarial data augmentation (DADA); Internet of vehicles (IoV); gameboy tile designer (GBTD); histogram of oriented gradie (HOG); graphene-poly lactic acid (GPLA); supervisory control and data acquisit (SCADA); deep neural network (DNN); multi-level marketing (MLM); Domain aware feature embedding (DAFE); domain adaptation (DA); machine learning (ML); state-of-the-art (SOTA); synthetic minority over-sampling technique (SMOTE); artificial neural networks (ANN); K-nearest neighbors (KNN); neural language processing (NLP).

4.1. Big data and good data

This section summarizes some of the most recent techniques used for assessing the quality of big data collected from various sources. The goal of applying these data-centric strategies is to improve the quality of big data and enhance the predictive capabilities of ML systems (Murphy, 2019). Highlighting whether big data are good or bad and presenting seven factors to determine the quality of data: validity, accuracy, relevance, granularity, consistency, accessibility, and security. The report indicated that instead of comparing good or bad data, it is more important to determine how big data are managed. Are big data managed efficiently enough to produce favorable outcomes for the business? The report further stressed the importance of cloud-based data management systems, ML, Hadoop, data- and text-mining techniques, and predictive analysis for the efficient use of big data. This report calls for a big data management process to separate wheat from chaff.

4.1.1. Big data

Tian et al. (2018) proposed a data quality assessment system based on data provenance theory to measure and monitor the quality of big theoretical data modeled on the power production management system (PMS) of the State Grid Corporation of China. The data provenance tracks the origin of the data. It has two main methods: the tagging method and the reverse query method. The proposed method tracks the source of the power data. Power data can be in the form of grid frequency, power-supply voltage, grid voltage, power-supply operation, power transmission, and transformation. It devises data checkout rules for six data characteristics: integrity, redundancy, timeliness, accuracy, consistency, and intelligence. The proposed data quality assessment and evaluation system improves the quality of the power data. This study is limited by the use of theoretical power data. The checkout rules may present different results for real-time power data.

Juneja and Das (2019) presented a comprehensive preprocessing framework to assess the quality of large amounts of data collected from conceptualized weather forecasting and monitoring applications. The authors based their framework on six intrinsic data-quality dimensions: uniqueness, timeliness, validity, completeness, accuracy, and consistency. The proposed model applied many preprocessing techniques, such as data integration, cleansing, and enrichment, and applied automated and user-defined domain filters to improve the quality of weather forecasting data. However, this work is limited by the lack of validation of real-time domain-specific data.

Taleb et al. (2018) developed a quality assessment and evaluation model to handle unstructured big data (UBD). Unstructured data have either unknown or no structure. The researchers analyzed the UBD life-cycle and highlighted the various data sources and methodologies used to examine them. The proposed data assessment model comprises seven steps for extracting quality data from UBD. These include identifying data quality requirements, identifying various data sources, managing data quality repositories, sampling and profiling UBD, and preparing textual, media, and Web data for quality evaluation, feature selection, and quality assessment reports. The proposed framework outlines an initial data

quality schema for UBD. Further experiments are required to develop a more robust UBD quality assessment model.

4.1.2. Good data

Taleb and Serhani (2017) presented a data quality rule discovery (DQR) framework to generate data quality (DQ) rules for big data before the preprocessing phase. A quality score was measured for each data feature based on data quality dimension (DQD) metrics, such as accuracy, completeness, and consistency. DQR generates a set of accurate preprocessing activity rules, e.g., deleting incomplete rows and removing or replacing missing attributes for each data feature based on its quality score before preprocessing. Once the quality rule was applied to the data feature, its quality scores were calculated again to perform further validation and quality optimization. Researchers have reported a systematic increase in the quality of big data features after the preprocessing stage based on samples from a large dataset. This research shares a DQR algorithm that can potentially be applied to big datasets, but does not highlight any future research directions. This research can be further validated by conducting experiments on different large datasets.

Explainable AI for training datasets was discussed by Anik and Bunt (2021). The authors explored data-centric explanations for ML systems. This is a new dimension for assessing the quality of a dataset. This study aims to explain the characteristics of training data to end-users in four different scenarios to record their reactions. Data-centric explanations allow end-users to trust the system and assess its fairness. Information on the data collection process, demographics, recommended usage, potential data issues, and other general information was recorded. Some questions were as follows: Who collected the data? What were the preprocessing steps? How were the data labeled? The data-centric explanations answer several questions. The researchers concluded that these explanations positively impacted end-users' sentiments regarding the ML system. However, the study was limited by hypothetical scenarios and a lack of objective data for end-user participants on a large scale.

Meng et al. (2020b) presented a comprehensive review of data fusion ML techniques. Data fusion is the process of combining multiple data sources to obtain accurate and consistent information. We added this survey paper to this section to highlight another important data-centric aspect of ML, which aims to gather high-quality datasets. This paper outlines seven data fusion challenges: data imperfection, inconsistency, conflict, alignment, data-type heterogeneity, fusion location, and dynamic fusion. The authors identified three data fusion types: signal, feature, and decision levels, and compared 30 previous studies that fell under these types. The study concludes by mentioning various open challenges, such as using DL techniques for data fusion and making data fusion models efficient, secure, and privacy-safe. However, these methods face the issues of computational cost and processing over a large dataset. Further research is required to explore data fusion techniques for large-scale systems.

4.2. Pre-processing techniques

Preprocessing is one of the most effective methods to enhance the

performance of AI models. The end goal of preprocessing is to improve the quality of the dataset. It involves various techniques, such as data collection, data augmentation, data or feature selection, data generation, data wrangling, data munging, and data labeling. The data can be improved by applying either one of these techniques or by combining multiple techniques. Collectively, these techniques present a data-centric approach to ML. Hence, the predictive capacity of the ML model can be significantly affected by performing high-quality data preprocessing. In this section, we review some modern data preprocessing techniques that have enabled researchers to develop novel models that outperform previous SOTA techniques.

4.2.1. Data collection & labeling

One of the most crucial aspects of ML is data collection and labeling. ML and DL models can benefit from using the various data collection and labeling approaches described in the literature in this section (Quan et al., 2020), addressing the problem of blind detection in computer-generated (CG) images. convolutional neural network (CNN)-based models were used to distinguish between real and computer-generated images. CG images were created using rendering techniques for which the model was not trained to provide a challenge for blind detection. The CG generalization problem is another name for this. Four CG picture datasets were generated using four distinct image-rendering tools employing data- and model-centric approaches. There is a difference between the data- and model-centric approaches in image production.

Additionally, a new method for identifying these photos was developed to learn numerous characteristics. The created images increase the generalization power of the detectors. The authors concluded that a new formulation for picture production could further boost generalization.

Wadekar et al. (2021) used a data-centric approach and offered a novel data-gathering strategy to address autonomous racing problems using DL techniques. According to a recent study, the DL prediction models for autonomous racing perform better when more data are collected. Researchers created a self-driving car simulator powered by Udacity on the Unity platform and employed a direct learning policy to increase the quality of the driving data collected by the simulator. Data aggregation and behavioral cloning are two parts of the direct learning policy. Based on expert feedback, both strategies increased the quality of the data generated during the simulation. A model for steering and throttle prediction was presented in this study. The model predictions were good.

Knowledge graph simple question answering (KGSQA) system (Sidiropoulos et al., 2020) suggested a RNN-based data-centric method for creating synthetic question-answer pairings on unknown domains. For this project, the authors devised a domain-adaptation framework and chose 6 target domains: science fiction, astronomy, literature, film, book and education. The new model for generating questions and answers outperformed the model-centric SOTA method in various contexts. However, researchers have determined that the question-generation process should be improved.

Kim et al. (2020) created techniques for simulating location-based social networks (LSBNs) data. Other social contacts, such as running, cycling, and other forms of physical activity (e.g., yoga or brisk walking), are also included in the LSBN. Previous LSBN datasets have been sparse and limited in size. They were also concerned about their privacy. These issues can be avoided by creating LSBN datasets synthetically. A geo-spatial support for multi-agent simulation of neighborhoods (GEO-MASON) was utilized by the researchers involved in this study to produce large-scale LSBN data based on lifestyle patterns (simulating how users behave naturally). Using the synthetic datasets generated by this technique, it is possible to benchmark the LSBN data for many research objectives, such as social connection prediction, location recommendation, and community discovery. However, this research was hampered by its

lack of experimentation in real-world situations. This will be examined further in future studies.

Bérard et al. (2020) built a multidomain and multilingual transformer-based language model to translate five languages into English, with particular focus on a biomedical dataset. The model is fluent in Korean, Italian, Spanish, German, and French. Domain tags and a single model enable translation between different domains. Tags were used to achieve more precise control over translation. This study produced a Korean-to-English translation dataset based on COVID-19 data for comparison. Compared to prior SOTA models, the innovative model presented in this research is a single model that can translate into many languages.

The biomedical training dataset was improved using a data-centric method by Yoon et al. (2022). The performance of the BioBERT language model on the NLP question-answering task was improved by removing noisy data and improving data labeling uniformity. The performance of the model improved as a result of the more accurate data. To support its findings, this study does not undertake much comparison and contrast.

Lee et al. (2019) presented a new data labeling method using a faster R-CNN deep learning algorithm to identify smoke and flames. A region proposal network (RPN) increases object detection in faster R-CNNs. The training images can be classified in two ways. There are two types of fire labels in a single system: huge fire labels and little labels. The other labeling systems included large, medium, and small categories. According to the authors, if a judicious labeling strategy is employed, the model performs well. For example, a small label may miss a large fire or smoke, whereas a large bounding box may miss a little fire. However, data labeling and analysis require further improvement.

Wang et al. (2020) developed an innovative system called LILA to generate synthetic data labels for incomplete and unlabeled data collections. Various data samples from real-world datasets frequently lack labeling information. It is common for only one class to be labeled and the other classes to be left out. There is a lack of coverage for all data types. The generator and discriminator frameworks of the generative adversarial network (GAN) were used in the proposed framework. The algorithm generates labels for unlabeled data points after learning the labeling patterns for already available classes. LILA was tested on various datasets to ensure accuracy of the results. To address the issue of multi-label class imbalance, synthetic label creation based on this architecture can be utilized in the future.

4.2.2. Data augmentation

Data augmentation improves the quality of the data by increasing its size. A modified version of the original data was added to the dataset. This represents an important data-centric aspect that must be highlighted further to emphasize its importance. In this section, we discuss various data augmentation techniques from the academic literature.

Zhang et al. (2020b) developed a deep adversarial data augmentation (DADA) technique based on a supervised GAN model. This paper proposes a new loss discriminator for the GAN model to generate high-quality images for problems with insufficiently labeled data. This study found that the DADA outperformed previous data augmentation techniques. The proposed technique was tested on three small real-world datasets to address classification problems. DADA produced favorable results. Furthermore, DADA was explored for object-detection tasks, which produced comparable results. The researchers concluded that DADA has extensive applications in object detection across various data domains.

Mansourifar et al. (2019) proposed a new concept of virtual big data to address the class imbalance problem by developing a GAN model called V-GAN to generate virtual big datasets. This oversampling technique produces high-dimensional data based on original training instances using the curse of dimensionality. It concatenated different

training data instances and created high-dimensional data points. The study conducted experiments on 10 different datasets and concluded that the proposed technique outperformed traditional GAN-based techniques in improving the classification prediction of imbalanced data.

Zhang et al. (2020a) presented a novel data augmentation technique called circular shift that performs data-independent augmentation operations to improve the classification accuracy of CNN-based models. Traditional data augmentation techniques, such as rotation, cropping, and flipping, are fairly good for improving classification accuracy. However, in the case of small images or dispersed features, these techniques can result in information loss and hence decrease the classification accuracy. The circular shift technique produces many transformed images while avoiding information loss. The proposed technique is tested on three datasets. These results are favorable for the CNN-based models. This technique can be improved in the future.

Chen et al. (2020b) proposed a residual condition generative adversarial network (RCGAN) that generates images of cervical single cells to improve the detection of cervical cancer. However, data on cervical cancer detection are limited. The RCGAN augmented the original data by generating high-quality samples. Using this technique, the cervical cancer dataset was expanded. The authors compared the classification accuracies of the proposed technique with those of previous techniques and observed an improvement in the classification accuracy of cervical cancer. The proposed RCGAN can be improved in the future to handle cell-edge information and generate better samples.

4.2.3. Feature engineering

The research from Cooney et al. (2021) conceptualized a neural network framework that adopts a data-centric approach to optimize the feed composition of aquatic life via a multicriteria analysis that leverages economic and nutritional lifecycle, and environmental datasets related to aquatic life. The framework aims to eliminate the trial-and-error system to move towards sustainable and optimized aquatic feeds. This could also help to lower the cost of the feed. The framework can perform extensive feature engineering to extract relevant features such as animal requirements, animal response, price, ingredients, and nutritional value to optimize feed requirements. The authors presented a theoretical framework in which practical implementation needs to be validated.

Sharma and Liu (2020) presented an ML-based data-centric misbehavior detection model for the Internet of vehicles (IoV). The proposed model is based on extracting quantitative features from vehicular datasets and applies two plausibility checks to the data, location and movement plausibility, which verifies the vehicle's location and movement. Six ML models were compared with the extracted features. Ensemble voting yields the best results. The researchers concluded that the proposed data-centric approach could be further validated using real-world datasets.

Punmiya and Choe (2019) presented a data-centric approach to energy theft detection. The proposed framework performs feature engineering during the preprocessing phase to introduce six new theft cases and four stochastic features. The synthesized features improve the performance of a gradient-boosting theft detector (GBTD), which consists of three techniques: XGBoost, CatBoost, and LightGBM. The results show that feature extraction reduced the dataset noise and improved the performance of GBTD. In addition, LightGBM improves the time complexity of theft detection, whereas LightGBM and CatBoost outperform XGBoost in terms of accuracy.

Huang et al. (2019) proposed a feature-engineering framework for predicting short-term earthquakes using acoustic and electromagnetism to AI (AETA) data and historical seismic event data. Five different feature-engineering techniques were used: (1) the sunrise-sunset (SRSS) auto-recognition algorithm, (2) the Higuchi fractal dimension method, (3) the sliding interquartile range anomaly method, (4) features from geo-sound precursor data, and (5) features from historical seismic event data. They concluded that the extracted features could be used to predict short-term earthquakes in the future.

Alzahrani et al. (2019) presented an inception V3 CNN model based on custom feature-engineering techniques to classify face shapes into 5 types: square, oval, heart-shaped, round, and oblong. Feature engineering techniques include a histogram of oriented gradients (HOG) paired with a support vector machine (SVM) linear classifier and features extracted from facial landmarks. The CNN classifier first extracts features based on the two feature extraction algorithms and then feeds on these features to classify face shapes. The comparative results show that the proposed technique performs better than the previous modern face shape detection techniques. However, this work can be further improved through better feature extraction and the use of different datasets.

4.3. Data management

This survey paper from Sanjeeva (2018) and Siddiqua et al. (2016) focused primarily on the big data aspects of data management. Therefore, a broad coverage of existing work on storage, preprocessing, processing, and security is provided. In addition, this survey offers added value through a comprehensive taxonomy of existing techniques and technologies. This highlights the importance of typical big data management challenges related to storage, preprocessing, processing, and security. Moreover, this survey aims to be a useful guide to challenges and solutions in big data management, and a reference point for future work on big data management. Furthermore, this survey summarizes the benefits that can be achieved if techniques are adopted for specific management applications, such as storage, preprocessing, analysis, and security. Most significantly, this study serves as a guide for researchers to explore suitable big data management techniques and develop augmented techniques in response to the insufficiency of existing solutions.

Data management (Schiermeier, 2018) is an example of public research sponsors and institutions implementing “open science”, pushing to make scientific research and data freely accessible. Over the past decade, many funding agencies have made data management plans mandatory for grant applicants. All the U.S. federal agencies, including the National Science Foundation and National Institutes of Health, have such policies. Data management plans must also be included in grant proposals to the European Research Council and other European Union-funded research programs. Many national funding agencies in Europe, including the U.K. Research Council and the London-based Wellcome Trust, the world's largest biomedical research charity, also ask for data plans.

Many scientists practice data management by default. For example, astronomers have been doing so for decades while calibrating their observations and archiving huge amounts of telescope survey data in standardized, machine-readable catalogs for reuse. According to Sundarraj (2022), “If you own or plan to start a business, you have probably got advice like ‘manage your data effectively’ and ‘focus on your customers more’ multiple times. However, suppose you believe there is excessive hype surrounding data management and customer relationships. In that case, it is time to learn how data management and customer satisfaction are related and how you can use them to help your business succeed.”

4.4. Semi-supervised learning

Semi-supervised learning has received significant attention from modern ML practitioners and researchers. It addresses one of the most important challenges faced by the AI community, namely, the non-availability of large, high-quality labeled datasets. This allows researchers to use small labeled datasets to train ML models and use these trained models to loosely label massive unlabeled datasets. Hence, semi-supervised learning combines supervised and unsupervised learning methods. In this section, we review the recent work of ML researchers and summarize various techniques based on semi-supervised learning, pseudo labeling, weak supervision, and active learning. Although these techniques differ in concept, they are all responsible for labeling

unlabeled datasets. In most scenarios, after applying these labeling techniques, the inclusion of unlabeled datasets in the training phase significantly improves the quality of the ML systems.

Bossé et al. (2020) experimented with various data-centric and model-centric approaches to improve the efficiency of artificial neural networks (ANN). They analyzed the impact of the incremental and cumulative training modes of the ANN. In the incremental training mode, active learning is applied to extract the most informative data, and the model is sequentially updated using the newly labeled data. The authors also explored semi-supervised deep active learning and developed a method called the gradual pseudo-labeling algorithm (GPLA) to annotate unlabeled data. The researchers concluded that the GPLA improved the test accuracy of the ANN in all the conducted experiments.

Chao et al. (2021) developed a deep neural network (DNN) based on small labeled wind turbine data collected via a supervisory control and data acquisition (SCADA) system. The trained model generated pseudo-labels for the unlabeled wind turbine data. Unlabeled data were converted into large amounts of labeled data. The experiments show that the pseudo-labeling technique can improve the DQ and model performance.

Shao et al. (2019) proposed a semi-supervised learning-based labeling technique for detecting malicious communications in advanced security threat detection. Labeled data in this field are scarce, and it is difficult to extract the relevant features. The proposed DNN-based technique can also learn the optimized feature representations and manage unknown labels. This technique shows an improvement in the labeling accuracy. The results showed that the DNN-based technique is more effective than the k-nearest neighbor (kNN) and label propagation algorithms (LPA) in terms of labeling accuracy. Meng et al. (2020) proposed a GAN-based deep semi-supervised learning model was proposed to generate labels for seismic impedance inversion. The GAN model was paired with a supervised CNN architecture to capture data features. Experiments showed that the GAN-based seismic labeling outperformed the regular CNN technique. However, this study may require additional experiments and comparisons with previous SOTA for a robust evaluation.

Noorbehbahani and Saberi (2020) proposed a novel approach to ransomware detection using semi-supervised learning. After careful experimentation, the authors selected the OneR or Chi-squared feature-selection techniques for the ransomware data. Random forest was used as the base classifier. The authors concluded that the proposed semi-supervised technique outperformed supervised ransomware detection. This research can be further improved by proposing semi-supervised feature-selection techniques. A novel technique for image classification was proposed in the study from Zhang and Yan (2020). It is based on cost-effective active learning (CEAL) and a tri-training algorithm collectively known as Tri-CEAL. Small-labeled samples were used to train Tri-CEAL using semi-supervised learning. It extracts high-confidence samples that are used for further labeling. The authors concluded that the Tri-CEAL model performed better on deep image classification tasks. They aim to achieve a higher labeling accuracy in the future.

4.5. Domain adaptation and transfer learning

Transfer learning has emerged as a vital ML technique to improve the quality of ML systems. This technique adopts a data-centric approach by fine-tuning a large number of pre-trained models. This transfers information from one ML model to another. This information includes pre-trained model weights and other model configurations. In this section, we review some of the latest transfer learning and domain-adaptation approaches.

Gururangan et al. (2020) investigated several pre-training variations to improve the performance of the RoBERTa language model. Researchers experimented with domain adaptive pre-training (DAPT) and tasked adaptive pre-training (TAPT). The following four domains were targeted: news articles, reviews, computer science publications, and

biomedical publications. The TAPT processed 8 classification tasks. They concluded that pre-training improved the performance of the RoBERTa model. In future work, we suggest devising more efficient domain-adaptation techniques and better data selection for TAPT.

Han and Eisenstein (2019) proposed an improved variant of the BERT model, known as AdaptaBERT, which performs fine-tuning to adapt to the unsupervised domain. Pretrained language models perform well for problems with similar task-specific datasets. However, the proposed model is highly effective when the pre-training and target data domains are completely different. AdaptaBERT follows a two-step domain-adaptive fine-tuning process: domain and task tuning. The model exhibited competitive results and outperformed other BERT variants in some cases. Future research directions include combining task-specific and domain tuning to achieve better results, and assessing the possibility of a single contextualized embedding model for different unsupervised domains.

Ben-David et al. (2020) presented a pivot-based encoder representation language (PERL) model for unsupervised DA, which is built on top of BERT and fine-tuned with a pivot-based masked language modeling (MLM) objective. The proposed model outperformed other language model variants on 22 binary sentiment classification tasks. In the future, we intend to extend this study to other NLP tasks.

Nguyen and Chang (2021) presented a data-centric approach to predict the severity of COVID-19, where relevant data are unavailable in abundance. The authors proposed a hybrid convolution-attention neural architecture with a transformer self-attention module and dense associative memory. The authors concluded that pre-training the proposed model improved its performance. Furthermore, the additional self-attention module captures global contextual information from the input data. In future research, the authors propose experimenting with different self-attention modules to achieve better results with the hybrid architecture.

Dou et al. (2019) proposed a method called domain-aware feature embedding (DAFE) based on unsupervised DA for neural machine-translation tasks. The experiments showed better results for machine translation. Furthermore, pairing the model with back translation further improved the results. In the future, different datasets and language pairs will be tested to validate these results.

4.6. State of MLOps

In this section, we analyze the recent works on MLOps. MLOps is a relatively new term in ML nomenclature. MLOps combine ML with deployment and delivery operations. It was born at the crossroads of DevOps, data engineering, and ML, and although the concept is similar to that of DevOps, its execution differs. ML systems are experimental in nature, with many components that are considerably more difficult to build and operate. MLOps are a set of methods used by data scientists and operation experts to collaborate and communicate. The use of these methods improves the quality of ML and deep-learning models, simplifies management processes, and automates their deployment in large-scale production contexts. The models can easily align with business demands and regulatory requirements. MLOps are gradually becoming stand-alone solutions for ML lifecycle management. Data collection, model generation (software development lifecycle and continuous integration/continuous delivery), orchestration, deployment, health, diagnostics, governance, and business KPIs fall under this umbrella.

Mäkinen et al. (2021) surveyed 331 data science experts from 63 countries to analyze the state of MLOps. The survey found that most companies were trying to determine how to use data effectively, whereas others were trying to develop more accurate models. However, few companies build and automate ML pipelines by following the practices of retraining and redeployment models.

Zhou et al. (2020) developed an ML pipeline platform using existing tools, namely Kubeflow, Gitea, and Drone, to check the feasibility of building a platform enabled by continuous integration and continuous

delivery (CI/CD). The platform verified the practicality of MLOps. They executed seven ML models and monitored their GPU time and resource consumption.

In the study from Fursin et al. (2020), an open-source CodeReef platform was developed, which enables cross-platform MLOps and supports diverse and continuously changing systems. These are referred to as MLSysOps. This can automate the deployment of ML pipelines. The platform includes various application programming interfaces (APIs) for handling ML tasks and a portable ML workflow. The platform shares reusable ML components and artifacts with customizable and reusable packages. It supports Linux, MacOS, Windows, Raspberry Pi, IOS, and Android. Because the platform is open-source, the ML community can expand its features.

Renggli et al. (2021) studied the impact of DQ dimensions in MLOps workflows. The authors highlighted 4 aspects of quality: (1) effective ML quality optimization for models, (2) prevention of unrealistic expectations from the dataset, (3) model testing to prevent overfitting, and (4) quality testing for data changes in the production environment.

The authors provided tools and solutions for each aspect to ensure quality and mentioned their limitations. They also highlighted future research directions regarding ML-aware data quality and testing of the MLOps workflow for metrics beyond accuracy, such as fairness, robustness, and explainability.

The work in Granlund et al. (2021) analyzed the possibility of integrating and scaling MLOps workflows for many organizations and presented two case studies highlighting the challenges faced. The solutions found in these case studies were largely applicable to these workflows. A more generic solution can be investigated in the future by identifying similar workflow patterns across organizations. Chao et al. (2021) introduced a system called submarine, which is a unified ML system to address challenges related to “ML tech debts”. The ML technology debt includes code, reproducibility, and portability issues. Other issues are related to infrastructure administration. The submarine was built using Kubernetes and Hadoop YARN to address administrative issues. It supports various ML frameworks to provide a streamlined production-ready ML workflow. It allows users to build ML models in a few lines of code, and offers templates to reduce the time spent running experiments. This is an easy-to-use platform for data scientists and ML infrastructure administrators.

Dilmegani (2021) discussed various MLOps tools and platforms. The following three major categories of MLOps, i.e., data management, modeling and operationalization, have been identified. The tools that fall into these categories collectively deliver end-to-end ML lifecycle management. Data management involves various data labeling and versioning tools. Modeling includes various tools for feature extraction, experimental tracking, and hyperparameter tuning. Operationalization includes various tools for model deployment and monitoring. Finally, the author lists the major MLOps tools that provide end-to-end ML lifecycle management, such as MLflow, H2O.ai, DataRobot, and Databricks. In the study of Miranda (2021), the landscape of MLOps was discussed in detail. The author describes the ML lifecycle as two-cycle looping. One loop governs the entire ML life cycle, whereas the inner loop governs the model life cycle. Some available tools apply only to the models, whereas others apply to more than one ML process. Cloud platforms, such as Amazon web services (AWS), Azure, and Google cloud platform (GCP), offer complete ML lifecycle tools. There are many other standalone MLOps tools, such as Neptune, Snorkel, Deepnote, Kubeflow, and Kedro.

4.6.1. Experiment tracking

In ML, experimental tracking is the process of saving all experiment-related information in every experiment. Broadly, experimental tracking is a component of MLOps, a larger ecosystem of tools and approaches dedicated to the operationalization of ML. ML teams implement experimental tracking in different ways, using spreadsheets, GitHub, or self-built platforms.

Czakov (2020) explained experimental tracking in detail. The author defined experimental tracking as the process of saving ML experimental metadata. This includes the environment configuration files, evaluation metrics, parameter configurations, and model weights. To save metadata, the experimental tracking system must have three components: (1) experiment database, (2) experiment dashboard, and (3) a client library that systematically displays all the data. The author has also differentiated between MLOps, experiment tracking, and ML model management, and terms experiment tracking a sub-task of MLOps, which is the focus of the model development phase. Finally, the author explained in detail how experimental tracking can be set up with ML pipelines and identified three methods: (1) managing metadata using a spreadsheet, (2) the GitHub file verification system, and (3) modern experimental tracking tools, such as Neptune. The authors concluded that experimental tracking tools are ideal for capturing ML metadata.

Schelter et al. (2017) proposed a lightweight platform to perform ML experimental tracking. The platform can extract and store ML metadata regarding the datasets, training epochs, models, hyperparameter tuning, and evaluation. In addition, the system can store common ML artifacts and automatically extract the ML pipeline metadata. The proposed system is integrated with popular ML frameworks, such as MXNet, SparkML, and scikit-learn. However, a limitation of this approach is that it cannot identify logical operations performed on the data and only focuses on physical operations.

Chen et al. (2020) discussed the advancements introduced in MLflow, an open-source tool for managing the ML lifecycle. This tool offers extensive ML experiment tracking, development, deployment, and reproducibility. One of these updates included the extraction of detailed insights from millions of ML experiments. In addition, MLflow provides APIs to export ML experimental data in tabular form. This paper discusses three features of MLflow including model registry, auto-logging tools, and experiment analytical functions. In the future, MLflow can be extended to improve data reproducibility, add model telemetry tools, and execute multistep workflows.

4.7. Can adding data hurt?

Nakkiran et al. (2021) showed that a variety of modern DL tasks exhibit a “double-descent” phenomenon where, as we increase model size, performance first gets worse and then gets better. Moreover, double descent occurred not only as a function of the model size but also as a function of the number of training epochs. The authors illustrated the above phenomena by developing a new complexity measure known as the effective model complexity and hypothesizing a generalized double descent based on this measure. Furthermore, model complexity enabled them to discover specific regimes in which increasing (or even quadrupling) the number of training samples degrades the test performance.

Adversarial training can produce robust models of perturbations on the input data, but usually at the expense of a decrease in standard accuracy. It is commonly believed that more training data will eventually help such adversarially robust models to generalize better to benign/unperturbed test data to mitigate this issue. However, in this study, we challenge this conventional belief and show that more training data can negatively affect the generalization of adversarially robust models in classification problems. They first investigated Gaussian mixture classification with linear loss and identified three regimes based on the strength of the adversary. More data improve the generalization of adversarially robust models in a weak adversary regime. In the medium adversary regime, with more training data, the generalization loss exhibited a double-descent curve, implying an intermediate stage in which more training data negatively affected the generalization. Finally, in a strong adversary regime, more data almost immediately cause the generalization error to increase. Subsequently, we analyzed a two-dimensional classification problem with a 0–1 loss. We proved that more data always hurt the generalization performance of adversarially

trained models with large perturbations. To complement their theoretical results, they conducted empirical studies on the Gaussian mixture classification, support vector machines (SVMs), and linear regression (Li et al., 2020; Min et al., 2021), and explained intuitively that the “more data hurt” stems from the “double descent” risk curve. As the model complexity increased, the prediction risk of the learned model first decreased, then increased, and then decreased again.

4.8. Repairing data

According to the study, data can be repaired by cleaning (Rekatsinas et al., 2017) using probabilistic reasoning. HoloClean has been introduced as a holistic system for restoring data. In HoloClean, data mending methods based on integrity requirements or external data sources are combined with quantitative methods based on the statistical features of the input data. HoloClean automatically developed a probabilistic program for data repair given an inconsistent dataset. Using theoretical breakthroughs in probabilistic inference, we present a set of improvements that ensure inference over HoloClean’s probabilistic model scales for cases with millions of tuples. A wide range of datasets exhibiting various types of faults was used to demonstrate that HoloClean is scalable and finds data fixes that are 90% accurate and 76% recallable, on average. This resulted in an average F1 improvement of more than twice that of the current best practices.

Data quality can be defined as the fitness of data to fulfil business requirements. This can be achieved through personnel, technology, and other processes (Trivedi et al., 2023). This ensured compliance and consistency, particularly when data from different databases were combined. Without proper DQ management, even minor errors can cause revenue loss, process inefficiency, and failure to comply with industry and government regulations (Lee et al., 2021). Thus, DQ and cleaning are always linked, as ensuring DQ is critical and necessary before the sharpening of the analytic focus can occur.

Data cleaning is the initial stage of any ML project and is one of the most critical processes in data analysis. This is a critical step to ensure that the dataset is devoid of incorrect data. It can be performed manually with data wrangling tools or completed automatically with a computer program. Data cleaning entails a slew of procedures that, once performed, make the data ready for analysis. Given its significance in numerous fields, there is a growing interest in developing efficient and effective data-cleaning frameworks. In this survey, some of the most recent advancements in data-cleaning approaches were examined for

their effectiveness, and future research directions were suggested to close the gap in each method (Bogner et al., 2021).

4.9. Data in deployment: a data-centric view of technical debt in AI

In this research paper from Sculley (2022), the author introduced the concept of technical debt in ML. The concept of technical debt originally came from the world of software engineering, where it has often been found that pushing to develop software very quickly can create long-term maintenance costs that must be paid back later, and that, if left unaddressed, can compound over time. Taking on debt is sometimes a useful strategy, but we should make careful choices about when and where to do so, and have a solid plan for how to pay it off later. The factors that create technical debt include breaking abstraction boundaries, poor documentation, and a lack of unit and system tests. Factors that can pay off technical debts include refactoring, improving documentation, and creating unit and integration tests. Surprisingly, we found that ML and AI systems have a special ability to create large amounts of technical debt. This is because ML systems have all the qualities of traditional software systems because they are built on code. However, they also have a second layer of issues because the behavior of these systems cannot be specified exactly in advance. Instead, their behavior is implicitly defined by the specific choices of model algorithms, hyperparameter settings, and training data used.

From the standpoint of data-centric AI, we can even make the case that the data are the largest potential source of technical debt in an ML system. This is true for two reasons. Fig. 2 shows all the components of a typical production-level ML system. The main idea behind depicting these blocks in the figure is to show that, in real-world ML systems, only a small fraction of the ML code is involved, and the surrounding infrastructure is vast, complex, and time-consuming to build. To build a successful ML application, resources must be invested into various components.

Zhang et al. (2022) further explained that there is a second, perhaps even more important, way to see that data can be a source of unexpectedly large technical debt. This is because the data define the behavior of our models and, in this way, take on the role of the code. Suppose that we want a vision model to perform well in identifying insects. In this example, whether it is an acoustic model for studying a vocal command or a movie recommendation model to assist a user in selecting entertainment for a Friday evening, the skills and behaviors of our machine are mostly determined by the training data we acquire and use (Jiang et al.,

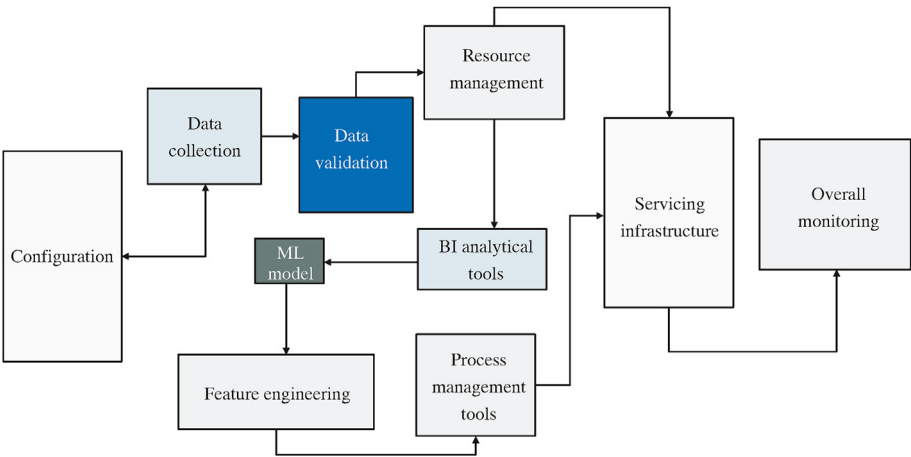


Fig. 2. Typical production-level machine learning system’s component.

2022). Why do experts in ML spend so much time, effort, and computing to obtain datasets containing billions of training examples?

This is because we want our models to perform well under a variety of conditions, and larger datasets are advantageous because of their greater diversity. This enables our models to learn more about a large trail of rare or exceptional conditions and determine how they should act within them; however, when all these unexpected occur, it can result in excessive, unnecessary costs that lead to debt.

5. Discussion and conclusion

In ML, researchers are exploring various techniques to improve the quality of their models (Abhishek and ud din Tahir, 2023). Data-centric modeling is an emerging aspect of ML. Although the term has been coined recently, ML practitioners have been using data-centric approaches for a long time. Data-centric modeling has emerged as a potent method for creating precise and dependable ML models. However, the data quality poses a serious challenge. For example, the data may be subject to incompleteness, noise, and bias, resulting in erroneous and unreliable models. However, the labeling of data is another essential part of data-centric modeling, which presents substantial obstacles such as time-consuming and error-prone labeling methods. Similarly, automated labeling methods and procedures can be used to overcome these obstacles, increase labeling accuracy, and decrease labeling time. Conventional algorithms for ML may not be able to manage the quantity and complexity of substantial datasets because of their inability to handle large data volumes and complexities. To effectively handle these obstacles, specialized techniques such as distributed learning and parallel processing can be applied. Data privacy and security are crucial concerns in data-centric modeling, particularly in fields such as healthcare and commerce, where private data are routinely exploited. Ethical and legal considerations must be addressed to secure personal data and to construct models in a responsible and moral manner. To address these concerns, suitable solutions such as data governance laws and privacy-preserving methods can be applied.

In this survey, we highlighted six major aspects of applying a data-centric approach to ML. The novelty of this survey lies in its comprehensive approach for presenting an end-to-end state of data-centric ML. In addition, the survey included a summary of recent SOTA data-centric ML techniques to guide future directions.

We believe that the most recent techniques for assessing big data quality can aid in improving the forecasting capabilities of ML systems. Therefore, the most effective technique for increasing the quality of datasets using an AI model was also explored in this survey (pre-processing technique).

Most scientists struggle with data collection and labeling, which are among the most important aspects of ML. As ML becomes a new type of software, data collection and quality improvement will become even more important for DL. We discuss four primary data activities (data collection, data labeling, data augmentation, and data management), which are all crucial in a data-centric strategy and have been explored by various communities but must be used together. This survey explained how the data were collected and labeled efficiently using various research techniques. Additionally, several data augmentation techniques from various academic studies were covered, as well as how data might be handled using data plans to avoid data misuse.

Unsupervised data are being developed at the same rate as supervised data during this era of exponential data growth. Consequently, semi-supervised learning has been used in a wide range of fields, including fintech, education, and entertainment. This survey introduced semi-supervised learning to increase the size of the necessary labeled datasets while learning from small amounts of labeled data. In addition, its dependable algorithm decreases the amount of annotated data utilized.

Domain adaptation and transfer learning were the other two techniques mentioned in this survey. Domain adaptation is a subset of transfer learning in which the two techniques function together. Domain

adaptation is a method that primarily uses deep neural networks to reduce domain shift or data distribution to improve the performance of the target task with the help of knowledge obtained from the source domain. This primarily addresses the situation in which we have different but related datasets for a common learning task. Transfer learning is equally useful for learning real-world tasks; however, it has limitations in most cases. Although deep DA has been successfully employed to solve various tasks, caution must be exercised when performing transfer learning, because brute-force transfer may harm the performance of our model.

The state of MLOps is discussed in the following section. Currently, the MLOps technology is constantly shifting. Furthermore, because each tool's reach may encompass various aspects of the MLOps process, each use case requires a thorough consideration of the MLOps tech stack.

Next, we addressed how adding data might harm the datasets and how HoloClean can be used to restore and clean them. Finally, we discuss the causes of technical debt in AI. Technical debt builds up when software design and implementation decisions run into "or outright collide with" business goals and timelines. This is why it can be found in almost every company: if you always wait until every line of code is flawless before deployment, your company will most likely not last long. However, when a company focuses excessively on deadlines, the team frequently cuts corners to achieve them, resulting in extremely large and unmanageable debt. This survey explains how unnecessary organizational debt is avoided in the literature review.

This review investigates technical concerns and solutions, as well as their legal foundations, to advance this subject in a manner that uses AI's great power to address real-world problems, while also considering the societal implications of these responses. Finally, we wish to emphasize that prejudices are deeply embedded in our culture, and it is a mistake to believe that AI and biased issues can be solved merely through technical solutions.

Nonetheless, because technology reflects and projects our biases into the future, it is the job of technology makers to recognize the limits of the technology and offer safeguards to avoid any hazards. Equally important is that technology makers recognize that technical solutions without a social or legal foundation cannot exist, necessitating multidisciplinary methods. In this era of data-centric AI, we believe that our study provides pertinent suggestions for practitioners.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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